

NOTE: This template is shareware downloaded from www.processimpact.com. All shareware payments are donated to the Norm Kerth Benefit Fund to help a consultant who is disabled with a brain injury. Please visit http://www.processimpact.com/norm_kerth.html to make a shareware payment (\$10 suggested). Thank you!

Software Requirements Specification

for

Honey's Cinema Version

Prepared by Seif Eldin Ahmed Awad 241015398
Youssef Tamer 241015374
Lojain Amr 241015469
Karma Koura 241015701

AAST

4/42026

eLearning versions of several popular Process Impact training seminars are available at www.processimpact.com/elearning.shtml, including "In Search of Excellent Requirements," "Exploring User Requirements with Use Cases," "Writing High-Quality Requirements," "Software Inspections and Peer Reviews," and "Project Management Best Practices". Single-user and corporate-wide site licenses are both available.

Table of Contents

Table of Contents	ii
Revision History	ii
1. Introduction.....	1
1.1 Purpose	1
1.2 Document Conventions.....	1
1.3 Intended Audience and Reading Suggestions.....	1
1.4 Project Scope	1
1.5 References.....	Error! Bookmark not defined.
2. Overall Description	2
2.1 Product Perspective	2
2.2 Product Features	2
2.3 User Classes and Characteristics	2
2.4 Operating Environment.....	2
2.5 Design and Implementation Constraints.....	2
2.6 User Documentation	3
2.7 Assumptions and Dependencies	3
3. System Features.....	3
3.1 System Feature 1	Error! Bookmark not defined.
3.2 System Feature 2 (and so on).....	Error! Bookmark not defined.
4. External Interface Requirements	6
4.1 User Interfaces	6
4.2 Hardware Interfaces	6
4.3 Software Interfaces	6
4.4 Communications Interfaces	7
5. Other Nonfunctional Requirements.....	7
5.1 Performance Requirements.....	7
5.2 Safety Requirements.....	7
5.3 Security Requirements.....	8
5.4 Software Quality Attributes.....	9
6. Other Requirements	Error! Bookmark not defined.
Appendix A: Glossary.....	Error! Bookmark not defined.
Appendix B: Analysis Models.....	Error! Bookmark not defined.
Appendix C: Issues List.....	Error! Bookmark not defined.

Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for the cinema including functional and non-functional requirements, constraints, and user expectations.

1.2 Document Conventions

This document is written using clear rules so anyone can read it

The following keywords are used in the requirement:

Shall is used for the requirements that must be implemented

Should is used for the recommended features

May is used for the optional features

The abbreviation are used:

FR: functional requirements

NFR: non functional requirements

UR: user requirements

SR: system requirements

1.3 Intended Audience and Reading Suggestions

This document is intended for developers, testers, and stakeholders. Developers should focus on Sections 3 and 4, while stakeholders should review Sections 1 and 2.

1.4 Project Scope

The scope of the Honey's Cinema Website is to let users to signup and login search and select movies, pick suitable dates and times, and reserve seats through a visual seating layout.

After booking, users can proceed to checkout and receive a digital ticket confirmation.

The system also includes an admin panel for cinema management to add, update, and remove movies, manage showtimes and cinema halls, and monitor user bookings.

2. Overall Description

2.1 Product Perspective

Honey's Cinema Website is a system developed to support the manual browse and ticket booking experience

2.2 Product Features

the major features of Honey's Cinema is to display movies with posters and titles, view detailed movie information including cast, crew and showtimes

Search moves by title

Select date and preferred showtime

Reserve seats using a visual seating layout

Complete ticket booking through a checkout page and receive a digital ticket

Admin panel for managing movies, cinema halls, schedules, and customer bookings

2.3 User Classes and Characteristics

User Class	Characteristics
Admin	Mange movies and schedules
Customers	Use the website to browse and select movies to watch
System Maintenance Engineers	They ensure the system remains available and stable
Cinema Manager	Cinema managers oversee cinema operations through the system

2.4 Operating Environment

The system is compatible with Google Chrome, Microsoft Edge, and Safari.

Users can access the website using all devices such as desktops, laptops, tablets, and smartphones.

The system requires a stable internet connection to browse movies, view showtimes, and complete ticket bookings.

Database: MySQL

2.5 Design and Implementation Constraints.

The system shall be developed with an HTML, CSS, and JavaScript front end and a PHP back end and comply with cybersecurity standards

The system shall use the existing company database

The system is planned to be completed within the academic project timeline specified by the university in week 13

2.6 User Documentation

The system may include a FAQ section and an online help center to assist users with common questions

2.7 Assumptions and Dependencies

It is assumed *users will have basic knowledge of using web browsers*

It is assumed users will have access to a stable internet connection

It is assumed data provided in the system will be accurate

It is assumed that the system will connect to a Payment gateway

3. System Features

Feature 3.1: User Signup and Login

Description: Allows users to securely sign up and log in using email and password.

Priority: High

Stimulus / Response Sequences

Stimulus	Response
User enters signup details	System creates a new account and stores it in the database
User enters correct login information	System allows access
User enters incorrect information	System displays an error message

Functional Requirements

- FR-1: The system shall provide a signup and login page
- FR-2: The system shall validate the email and password
- FR-3: The system shall allow access after verifying the email and password

- FR-4: The system shall store user account information securely in the database.

Feature 3.2: Movie Search

Description: Allows users to search by movie name

Priority: High

Stimulus / Response Sequences

Stimulus	Response
User enters 3 or more characters in search bar	System displays matching movie results
User enters less than 3 characters	System displays an error message
User enters unknow movie	System displays "No movies found"

Functional Requirements

- FR-1: The system shall provide a search par
- FR-2: The system shall the user to search by move name
- FR-3: The system shall only accept search input of three characters or more
- FR-4: The system shall display matching movies
- FR-5: The system shall show a “No results found” message when no matches exist

Feature 3.3: Date and Showtime Reservation

Description: Allows users to select a movie, choose a date and showtime, and reserve available seats

Priority: High

Stimulus / Response Sequences

Stimulus	Response
User click on a movie	System opens the movie details page and displays full information
User selects an available date and showtime	System displays the visual seating layout for the selected showtime
User selects an available seat	System marks the seat as temporarily reserved and tarn it gray and add it to the total

Functional Requirements

- **FR-1:** The system shall allow users to select a movie, date, and showtime.
- **FR-2:** The system shall display the movie poster, cast and crew on the movie details page
- **FR-3:** The system shall display the ticket price for the selected movie
- **FR-4:** The system shall display available seats using a visual seating layout.
- **FR-5:** The system shall prevent users from selecting red seats
- **FR-6:** The system shall update seat status when a seat is selected.
- **FR-7:** The system shall update the total every time seat is selected

Feature 3.4: Checkout

Description: Allow user to checkout using online payment

Priority: High

Stimulus / Response Sequences

Stimulus	Response
User click on a checkout	System opens the checkout page
User enters card information	System check if the information is correct
User enters on pay	System processes the payment
Payment is successful	System generates a digital ticket and tern the seats to red

Functional Requirements

FR-1: The system shall provide a checkout page for payment processing.

FR-2: The system shall validate payment information entered by the user

FR-4: The system shall confirm the booking after successful payment.

FR-5: The system shall generate a digital ticket after successful checkout.

Feature 3.5: Admin Management

Description: Allow admin to add and remove a move

Priority: High

Stimulus / Response Sequences

Stimulus	Response
Admin enters movie information	System add movie to database
Admin click on remove	System remove movies from the database
Admin updates movie information	System saves the updated movie details
Admin updates hall information	System saves the updated details

Functional Requirements

FR-1: The system shall allow admin to add movie by name, author, time of release cast and crew

FR-2: The system shall allow admin to remove movies

FR-3: The system shall allow the admin to update existing movie information.

FR-4: The system shall allow the admin to update hall information

FR-5: The system shall show a successful message

4. External Interface Requirements

4.1 User Interfaces

Customer Interface

- **Homepage:** Displays currently showing movies with posters and titles
- **Movie Details Page:** Shows full movie information including cast, showtimes, and ticket price
- **Booking Page:** Allows users to select showtime, choose seats, and confirm ticket reservations
- **Login/Signup Page:** Enables users to create accounts or log in
- **Payment Page:** Allows secure online payment using available payment methods

Admin Interface

- **Admin Dashboard:** Displays system overview and quick access to management functions
- **Movie Management Page:** Allows admins to add, update, and remove movies.
- **Showtime Management Page:** Enables scheduling and editing movie showtimes

4.2 Hardware Interfaces

- Customers can access Honey's Cinema Website using desktops, laptops, tablets, or smartphones with an internet connection and a web browser.
- Administrators use a desktop or laptop with a stable internet connection, keyboard, and mouse to manage movies and customer bookings.

4.3 Software Interfaces

- The system uses a database to store customer accounts, movie details, booking information, and payment records securely

- The system can run on common operating systems such as Windows, Linux, and macOS for both users and administrators
- Users can access the website through web browsers such as Google Chrome and Microsoft Edge
- The system provides user authentication through login and registration features using email and password verification
- The system supports secure online payment methods for ticket booking, such as credit cards and digital payment services
- The website will connect to a Payment gateway to handle tickets

4.4 Communications Interfaces

The system shall communicate with the database and external services using HTTPS to ensure secure data transmission. All data exchanged between the website, server, and database shall be formatted using JSON for consistency and easy processing.

Payment Gateway Interface: The system shall communicate with a **Bank/Payment Gateway API** to process ticket transactions

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- NFR-1: The system shall load pages and display movie information within 3 seconds under normal internet conditions.
Reference: [Google Search Central - Core Web Vitals](#) (Recommended LCP is 2.5s or less)
- NFR-2: The system shall handle multiple users booking tickets at the same time without performance degradation.
- NFR-3: The system shall process seat selection and checkout operations less than **500ms** to ensure a smooth user experience.
Reference: [Nielsen Norman Group - Response Times](#) (0.1s is instant, 1.0s is the limit for a seamless flow)
- NFR-4: The system shall synchronize seat status across all active sessions within **1 second** of a selection to avoid double-booking errors.
Reference: [AWS Whitepapers - Performance Efficiency](#) (Standard for real-time data consistency in distributed systems)

5.2 Safety Requirements

- NFR-1: The system must save **100% of all bookings** and payments immediately to the database so that no information is lost if the website crashes
Reference: [MySQL Reliability Standards](#)
- NFR-2: The system must update seat availability in **less than 1 second** so that two people cannot buy the same seat at the same time
Reference: [W3C Web Consistency Guide](#)
- NFR-3: The system shall provide confirmation messages before deleting or updating critical data such as movies or bookings.

5.3 Security Requirements

- NFR-1: The system must use a secure login process that checks 100% of emails and passwords against the database before allowing entry.
Reference: [Google Login Security Standards](#)
- NFR-2: The system must hide 100% of sensitive data, like passwords and credit card numbers, so they are unreadable to hackers
Reference: [OWASP Data Encryption Guide](#)
- NFR-3: The system must block **100% of normal users** from touching the Admin features, ensuring only authorized staff can manage movies
Reference: [IBM Access Control Basics](#)
- NFR-4: The system must use **HTTPS for 100% of communication** between the user's phone/computer and the server to keep data safe
Reference: [DigiCert HTTPS Guide](#).

5.4 Software Quality Attributes

- NFR-1: The interface shall be simple enough that new users can complete a ticket booking in under 3 minutes without needing a help guide.

Reference: [Nielsen Norman Group - Usability 101](#)

- NFR-2: The system shall ensure 99.9% availability

Reference: [Atlassian - Service Level Objectives \(SLOs\)](#)

- NFR-3: The system shall work perfectly on 100% of the latest versions of Google Chrome, Microsoft Edge, and Safari

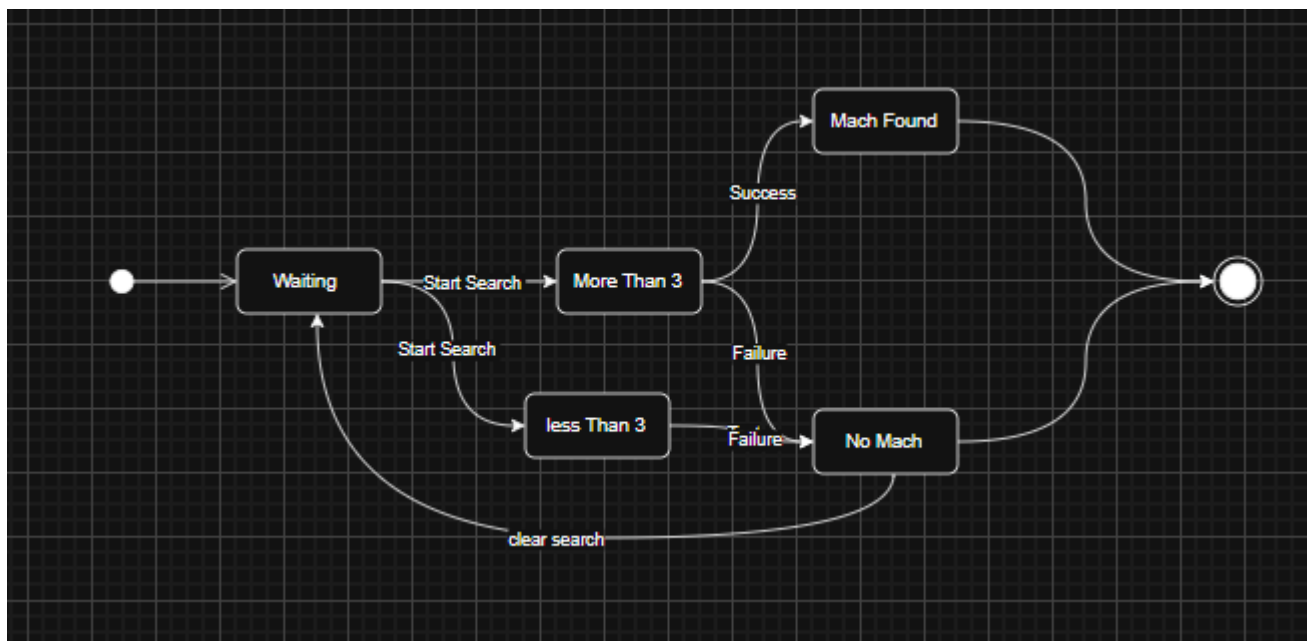
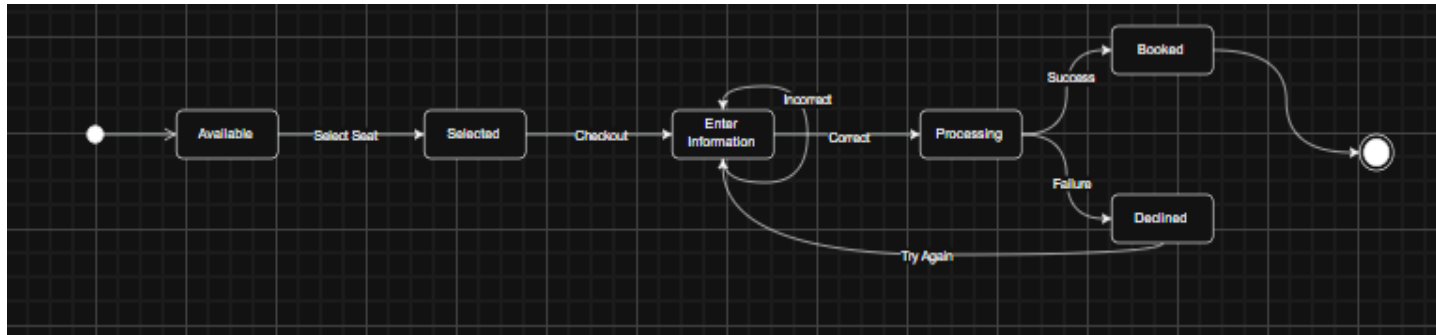
- **Reference:** [W3C Web Standards](#)

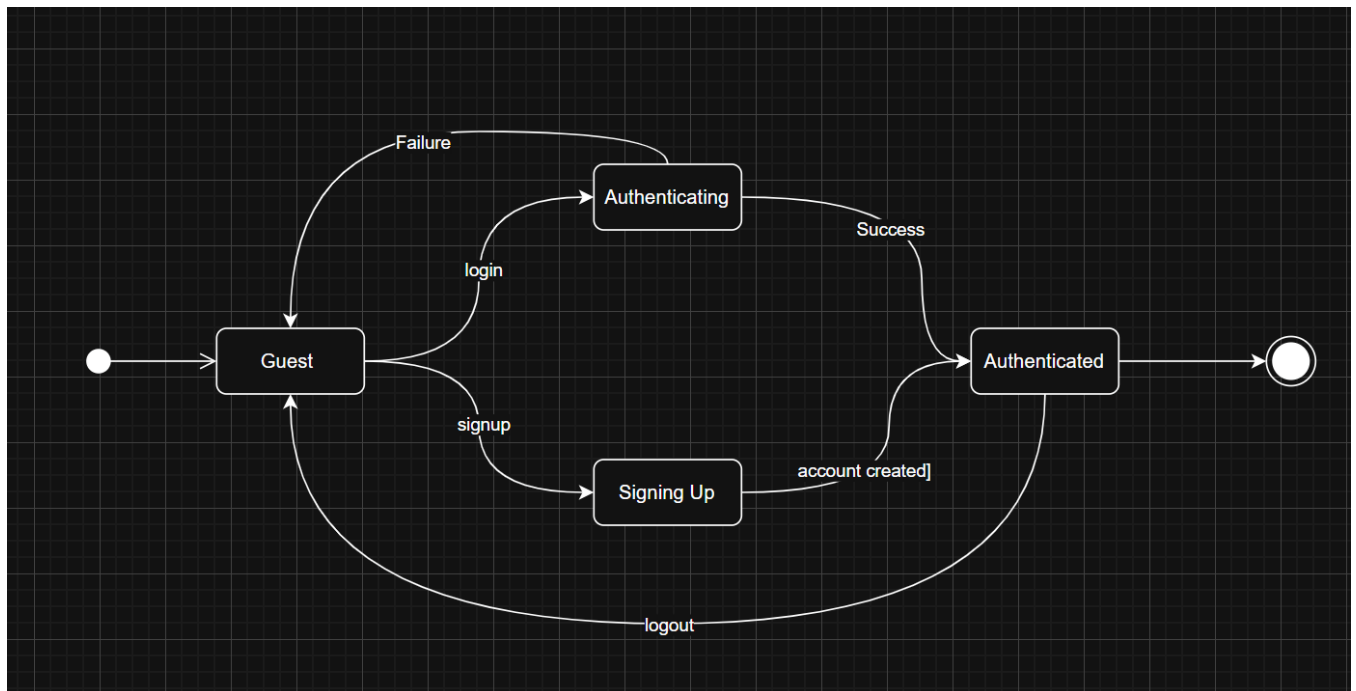
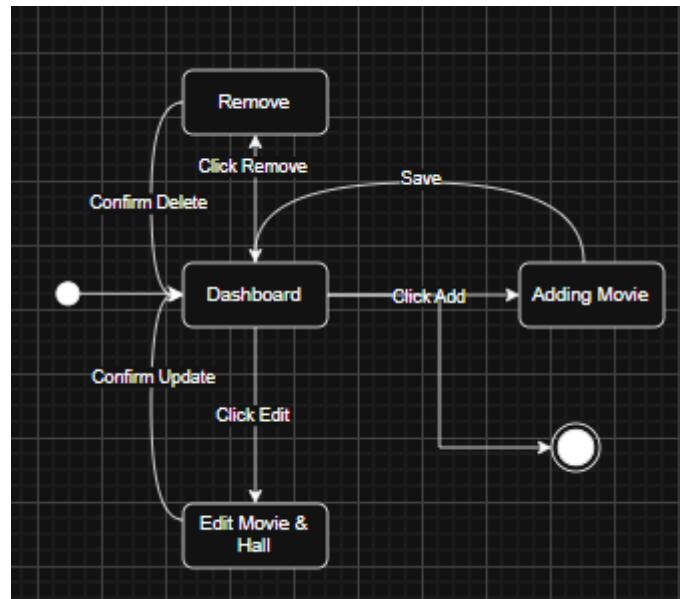
- NFR-4: The website must automatically resize and function correctly on **100% of screen sizes**

Reference: [Google - Responsive Web Design Basics](#).

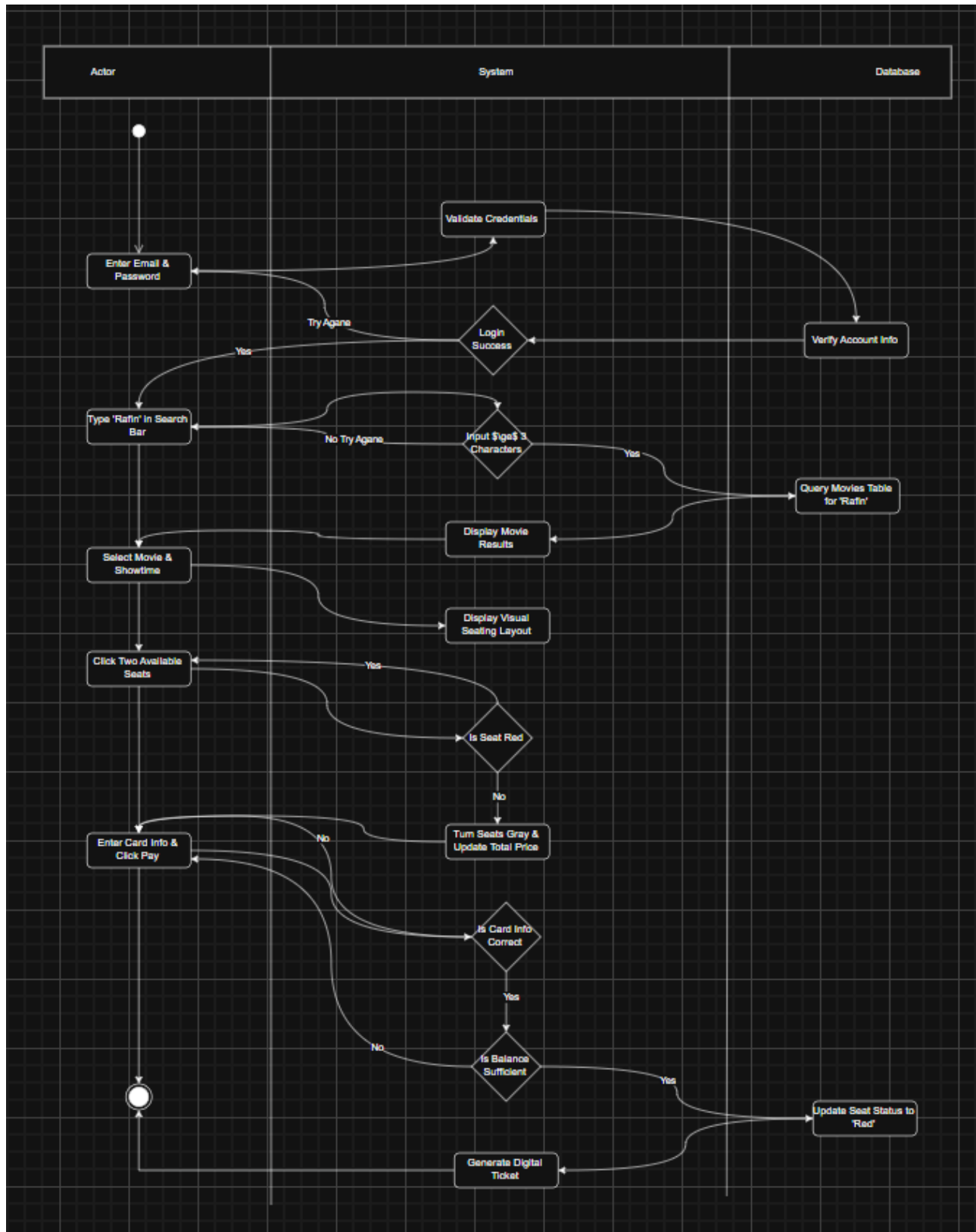
- NFR-5: The system shall be maintainable and allow future updates without major changes to the system structure

6.1 State Diagram

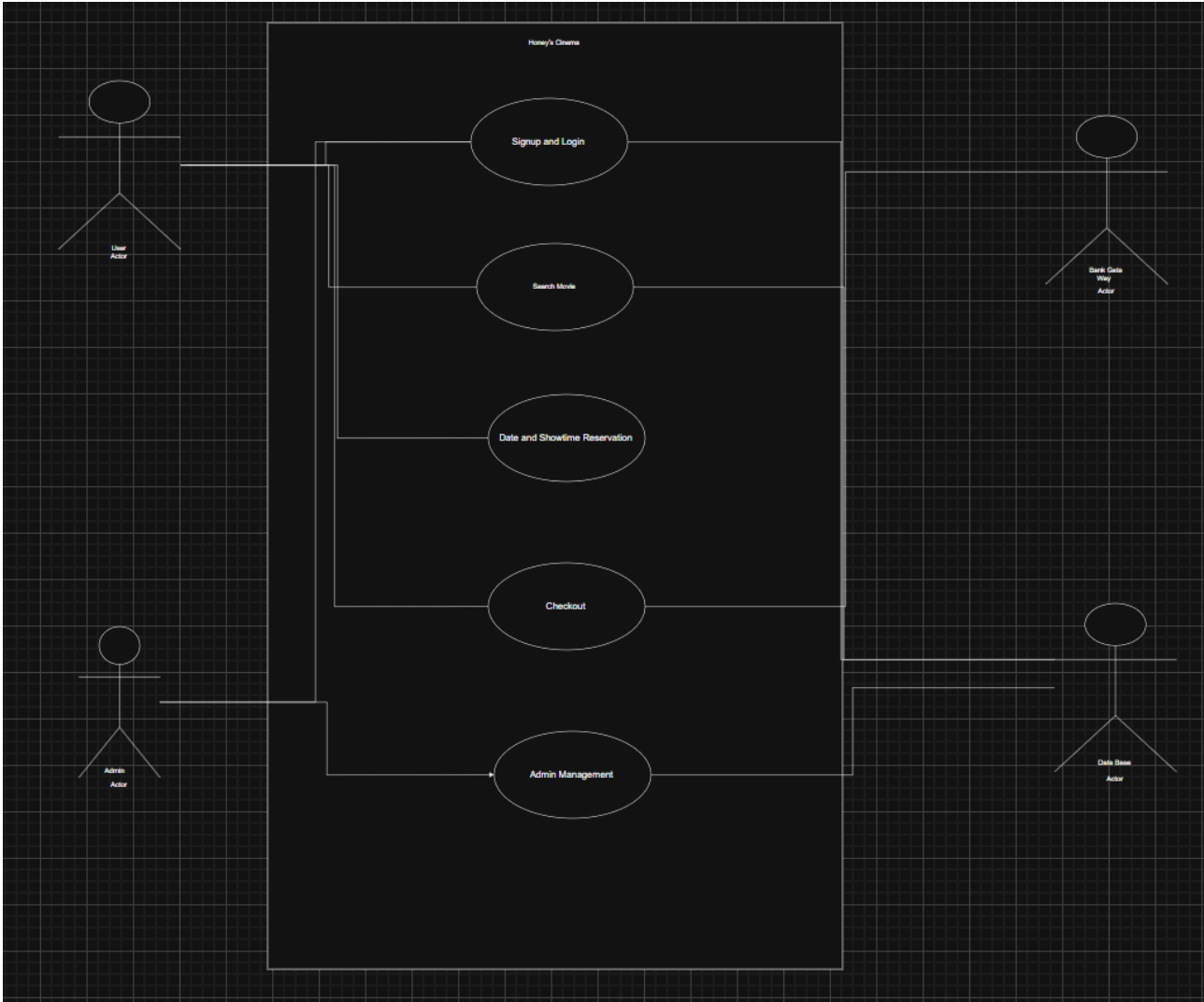




6.2 Activity Diagram



6.3 Use Case



6.4 Class Diagram

